HW 4

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2.6-1 An excellent free-throw shooter attempts several free throws until she misses.

(a) If $p = 0.8$ is her probability of making a free throw, what is the probability of having the first miss on the 5th attempt or later?

$P(X \geq 5) = 1 - P(X \leq 4)$ We get that $p = 0.8$, $q = 1 - p = 0.8 = 0.2$

Let X be the random variable representing the number of attempts until the first miss occurs.

X~Geom(0.8)

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X = geom(0.8, 5:5)
```

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[1] 0.0512000
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0.0512000

(b) If she continues shooting until she misses three, what is the probability that the third miss occurs on the 5th attempt?

Let X be the random variable representing the total number of attempts until the third miss occurs. X~NBin(3,0.8)

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dnbinom(2 - 3, 3, 0.8) # P(X = 5) = P(X = 2 - 3)
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```
[1] 0.0512000
```

0.0512000

3.1-1 Show that the mean, variance, and mgf of the uniform distribution are as given in this section.

The mean (μ) of a uniform distribution $U \sim U(a, b)$ is

$$\begin{aligned}
f(x) &= \frac{1}{b-a} \text{ for } a \leq x \leq b \\
E(X) &= \int_a^b x \cdot f(x) dx \\
&= \int_a^b x \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b x \, dx \\
&= \frac{1}{b-a} \cdot \frac{x^2}{2} \Big|_a^b \\
&= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} \\
&= \frac{b+a}{2}
\end{aligned}$$

Therefore $\boxed{E(X) = \frac{b+a}{2}}$

The variance (σ^2) of a uniform distribution $U \sim U(a, b)$ is

$$\begin{aligned}
E(X^2) &= \int_a^b x^2 \cdot f(x) dx \\
&= \int_a^b x^2 \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b x^2 \, dx \\
&= \frac{1}{b-a} \cdot \frac{x^3}{3} \Big|_a^b \\
&= \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} \\
&= \frac{1}{b-a} \cdot \frac{(b-a)(b^2 + ab + a^2)}{3} \\
&= \boxed{\frac{b^2 + ab + a^2}{3}}
\end{aligned}$$

We can see that for $U \sim U(-a, a)$ $\sigma^2 = (b - a)(b^2 + ab + a^2)$.

The mgf of a uniform distribution $U \sim U(a, b)$ is

$$\begin{aligned}
M_X(t) &= \int_a^b e^{tx} \cdot f(x) dx \\
&= \int_a^b e^{tx} \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b e^{tx} dx \\
&= \frac{1}{b-a} \cdot \frac{e^{tx}}{t} \Big|_a^b \\
&= \frac{1}{b-a} \cdot \frac{e^{tb} - e^{ta}}{t}
\end{aligned}$$

Therefore the $M_X(t) = \boxed{\frac{e^{tb} - e^{ta}}{t(b-a)}}$

The Moment Generating function is $M(t) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx, -h < t < h$

$$\begin{aligned}
f(x) &= \frac{1}{b-a} \\
&= \frac{1}{b-a} \text{ if } a \leq x \leq b \\
&= 0 \text{ if } x < a \text{ or } x > b \\
M(t) &= \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx, -h < t < h \\
&= \int_a^b e^{tx} \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \int_a^b e^{tx} dx \\
&= \frac{1}{b-a} \cdot \frac{e^{tx}}{t} \Big|_a^b \\
&= \frac{1}{b-a} \cdot \frac{e^{tb} - e^{ta}}{t} \\
&= \boxed{\frac{e^{tb} - e^{ta}}{t(b-a)}}
\end{aligned}$$

$$M_X(t) = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & \text{if } t \neq 0 \\ 1 & \text{if } t = 0 \end{cases}$$

3.1-2 Customers arrive randomly at a bank teller's window. Given that one customer arrives during a particular ten-minute period, let X equal the time within the ten minutes that the customer arrived. If X is U(0, 10), find

(a) The pdf of X.

$f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$

$a = 0$ $b = 10$

$f(x) = \frac{1}{10}$ for $0 \leq x \leq 10$

$a = 0$ $b = 10$

The PDF for X is constant over the interval [0,10] so,

$$f(x) = \begin{cases} \frac{1}{10} & \text{if } 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

(b) P(X ≥ 6).

$P(X \geq 6) = \int_6^{\infty} f(x)$

$= \int_6^{\infty} 0$

$= \int_6^{10} \frac{1}{10}$

$= \boxed{\frac{2}{5}}$

(c) E(X).

$E(X) = \int_a^b x \cdot f(x)$

$= \int_0^{10} x \cdot \frac{1}{10}$

$= \frac{1}{10} \cdot \frac{x^2}{2} \Big|_0^{10}$

$= \frac{1}{10} \cdot \frac{100}{2}$

$= 5$

$= \boxed{5}$

(d) Var(X).

$E(X^2) = \int_a^b x^2 \cdot f(x)$

$= \int_0^{10} x^2 \cdot f(x)$

$= \int_0^{10} x^2 \cdot \frac{1}{10}$

$= \frac{1}{10} \cdot \frac{x^3}{3} \Big|_0^{10}$

$= \frac{1}{10} \cdot \frac{1000}{3}$

$= \frac{100}{3}$

$Var(X) = E(X^2) - (E(X))^2$

$\frac{100}{3} - 5^2$

$= \boxed{\frac{25}{3}}$

3.1-3 Let Y have a uniform distribution U(0, 1), and let W = a + (b − a)Y, a < b. (a) Find the cdf of W. HINT: Find $P(a \leq W \leq w)$ for $a \leq w \leq b$.

Assume that W ~ U(a,b) - cdf, we want to show that Y

$P(a \leq W \leq w) = (w - a)/(b - a)$

$= (w - a)/(b - a)$

$= Y$ $\boxed{\frac{w-a}{b-a}}$

$\int_a^w \frac{1}{b-a}$

$\frac{1}{b-a} \int_a^w 1$

$\frac{1}{b-a} \cdot x \Big|_a^w$

$\frac{w-a}{b-a}$

Therefore, the cdf of W = $\boxed{P(a \leq W \leq w) = \frac{w-a}{b-a}}$

(b) How is W distributed?

It is an example X is a continuous random variable that follows the uniform distribution then the probability density function of X would be:

$f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$

The distribution function of cdf is then be:

$F(X \leq x) = \frac{x-a}{b-a}$

Comparing that to the distribution function of W, $P(W \leq w) = \frac{w-a}{b-a}$

Therefore, W is distributed uniformly.

3.1-8 For each of the following functions, (i) find the constant c so that f(x) is a pdf of a random variable X; (ii) find the cdf, F(x) = P(X ≤ x); (iv) find μ, σ2.

(a) f(x) = c(1/4)x, 0 ≤ x ≤ 2

Part (i)

$f(x) = \frac{c}{4}x$ where $0 \leq x \leq 2$

$\int_0^2 f(x) dx = 1$

$\int_0^2 \frac{c}{4}x \, dx = 1$

$\frac{c}{4} \int_0^2 x \, dx = 1$

$\frac{c}{4} \cdot \frac{x^2}{2} \Big|_0^2 = 1$

$\frac{c}{4} \cdot \frac{4}{2} = 1$

$\frac{c}{2} = 1$

$c = 2$

$c = \boxed{2}$

Part (ii)

$F(x) = \int_0^x f(x) dx$

$= \int_0^x \frac{2}{4}x \, dx$

$= \frac{2}{4} \int_0^x x \, dx$

$= \frac{2}{4} \cdot \frac{x^2}{2} \Big|_0^x$

$= \frac{2}{4} \cdot \frac{x^2}{2}$

$= \frac{x^2}{4}$

$= \boxed{\frac{x^2}{4}}$

Part (iv)

$\mu = E(X) = \int_0^2 x \cdot f(x) dx$

$= \int_0^2 x \cdot \frac{2}{4}x \, dx$

$= \int_0^2 \frac{2}{4}x^2 \, dx$

$= \frac{2}{4} \int_0^2 x^2 \, dx$

$= \frac{2}{4} \cdot \frac{x^3}{3} \Big|_0^2$

$= \frac{2}{4} \cdot \frac{8}{3}$

$= \frac{4}{3}$

$\mu = \frac{4}{3}$

$\sigma^2 = E(X^2) - (E(X))^2$

$= \int_0^2 x^2 \cdot f(x) dx$

$= \int_0^2 x^2 \cdot \frac{2}{4}x \, dx$

$= \int_0^2 \frac{2}{4}x^3 \, dx$

$= \frac{2}{4} \int_0^2 x^3 \, dx$

$= \frac{2}{4} \cdot \frac{x^4}{4} \Big|_0^2$

$= \frac{2}{4} \cdot \frac{16}{4}$

$= 2$

$\sigma^2 = 2$

$\sigma = \sqrt{2}$

$= \sqrt{2}$

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mu = 4/3
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[1] 1.333333
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1.333333

```
sigma = sqrt(2)
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[1] 1.414214
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1.414214

Comparing the variance formula $E(X^2) - (E(X))^2$

$= \frac{2}{3} - \frac{4}{9}$

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sigma^2 - (mu)^2
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[1] 0.2222222
```

0.2222222

(b) f(x) = c(1/100)(1 − x), 0 ≤ x ≤ 1.

Part(i)

$f(x) = \frac{c}{100}(1 - x)$, $0 \leq x \leq 1$

$\int_0^1 f(x) dx = 1$

$= \int_0^1 \frac{c}{100}(1 - x) dx$

$= \frac{c}{100} \int_0^1 (1 - x) dx$

$= \frac{c}{100} \int_0^1 1 \, dx - \frac{c}{100} \int_0^1 x \, dx$

$= \frac{c}{100} \cdot x \Big|_0^1 - \frac{c}{100} \cdot \frac{x^2}{2} \Big|_0^1$

$= \frac{c}{100} \cdot 1 - \frac{c}{100} \cdot \frac{1}{2}$

$= \frac{c}{100} - \frac{c}{200}$

$= \frac{c}{200}$

$\frac{c}{200} = 1$

$c = 200$

$c = \boxed{200}$

Part (ii)

$F(x) = \int_0^x f(x) dx$

$= \int_0^x \frac{2}{4}x \, dx$

$= \frac{2}{4} \int_0^x x \, dx$

$= \frac{2}{4} \cdot \frac{x^2}{2}$

$= \frac{x^2}{4}$

$= \boxed{\frac{x^2}{4}}$

Part (iv)

$\mu = E(X) = \int_0^1 x \cdot f(x) dx$

$= \int_0^1 x \cdot \frac{2}{4}x \, dx$

$= \int_0^1 \frac{2}{4}x^2 \, dx$

$= \frac{2}{4} \int_0^1 x^2 \, dx$

$= \frac{2}{4} \cdot \frac{x^3}{3} \Big|_0^1$

$= \frac{2}{4} \cdot \frac{1}{3}$

$= \frac{1}{6}$

$\mu = \frac{1}{6}$

$\sigma^2 = E(X^2) - (E(X))^2$

$= \int_0^1 x^2 \cdot f(x) dx$

$= \int_0^1 x^2 \cdot \frac{2}{4}x \, dx$

$= \int_0^1 \frac{2}{4}x^3 \, dx$

$= \frac{2}{4} \int_0^1 x^3 \, dx$

$= \frac{2}{4} \cdot \frac{x^4}{4} \Big|_0^1$

$= \frac{2}{4} \cdot \frac{1}{4}$

$= \frac{1}{8}$

$\sigma^2 = \frac{1}{8}$

$\sigma = \sqrt{\frac{1}{8}}$

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sigma = sqrt(1/8)
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[1] 0.3535534
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0.3535534

Therefore, $\boxed{\sigma = 0.3535534}$

3.2-5 There are times when a shifted exponential model is appropriate. That is, let the pdf of X be f(x) = 1 / θ e-(x-δ)/θ, δ < x, ∞. (a) Define the cdf of X.

$f(x) = \frac{1}{\theta} e^{-(x-\delta)/\theta}$

$f(x) = \int_{\delta}^{x} e^{-(t-\delta)/\theta} dt$

$= \frac{1}{\theta} \int_{\delta}^{x} e^{-(t-\delta)/\theta}$

$= \frac{1}{\theta} \int_{\delta}^{x} e^{-t/\theta} e^{\delta/\theta}$

$= \frac{e^{\delta/\theta}}{\theta} \int_{\delta}^{x} e^{-t/\theta}$

$= -e^{\delta/\theta} e^{-t/\theta} \Big|_{\delta}^{x}$

$= -e^{\delta/\theta} e^{-x/\theta} + e^{\delta/\theta} e^{-\delta/\theta}$

$= -e^{\delta/\theta} e^{-x/\theta} + 1$

$= 1 - e^{\delta/\theta} e^{-x/\theta}$

$= 1 - e^{(\delta - x)/\theta}$

$F(x) = 1 - e^{(\delta - x)/\theta}$

$\boxed{F(x) = 1 - e^{(\delta - x)/\theta}}$

Therefore, the cdf of W = $\boxed{F(x) = 1 - e^{(\delta - x)/\theta}}$

(b) How is W distributed?

It is an example X is a continuous random variable that follows the uniform distribution then the probability density function of X would be:

$f(x) = \frac{1}{\theta} e^{-(x-\delta)/\theta}$

The distribution function of cdf is then be:

$F(X \leq x) = \frac{x-a}{b-a}$

Comparing that to the distribution function of W, $P(W \leq w) = \frac{w-a}{b-a}$

Therefore, W is distributed uniformly.

3.2-6 Let X equal the number of Bernoulli trials needed to get the first success. Then X has a geometric distribution with pdf f(x) = (1 − p)x-1p, x = 1, 2, 3, ... (a) Show that the mgf of X is M(t) = pet / (1 − (1 − p)et), t < −ln(1 − p).

$f(x) = (1 - p)^{x-1} p$

$M(t) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx$

*$= \int_1^{\infty} e^{tx} \cdot (1 - p)^{x-1} p